with $\zeta = \left[\mathrm{k_B T}\left(1 - e^{-2\gamma h}\right)\right]^{1/2}$, and $\mathbf{R}_n$ a vector of unit normal random numbers. We can write this update scheme succinctly as

$$\begin{aligned}\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_n,\\ \boldsymbol{p}_{n+1} &= e^{-h\gamma}\boldsymbol{p}_n - he^{-h\gamma}\nabla U(\boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_n) + \zeta\boldsymbol{M}^{1/2}\mathbf{R}_n.\end{aligned}$$

In the case of the one-dimensional harmonic oscillator, we have $U'(q) = \Omega^2 q$, yielding the compact form of one iteration as

$$\begin{bmatrix} q_{n+1} \\ p_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & h/m \\ -\Omega^2 h e^{-h\gamma} & (1 - \Omega^2 h^2/m)e^{-h\gamma} \end{bmatrix} \begin{bmatrix} q_n \\ p_n \end{bmatrix} + \begin{bmatrix} 0 \\ \zeta m^{1/2}\mathrm{R}_n \end{bmatrix}. \tag{7.12}$$

Taking expectations of (7.12) and then taking the limit as $n \to \infty$, we see immediately that

$$\langle q \rangle_h^{[\![\mathrm{ABO}]\!]} = \langle p \rangle_h^{[\![\mathrm{ABO}]\!]} = 0,$$

as $\mathbb{E}(\mathrm{R}_n) = 0$. Plugging the values for ψ_{ij} into (7.11), with

$$\bar{\mu}_{11} = \bar{\mu}_{12} = 0, \quad \bar{\mu}_{22} = m\zeta^2,$$

we can solve to find the exact value of the method's stationary averages. For example, the relative increase in the variance of position is computed for the $[\![\mathrm{ABO}]\!]$ scheme as

$$\begin{aligned}\frac{\langle q^2 \rangle_h^{[\![\mathrm{ABO}]\!]}}{\langle q^2 \rangle} &= \frac{e^{h\gamma} m(1 + e^{-h\gamma})^2}{2m(1 + e^{-h\gamma}) - h^2\Omega^2 e^{-h\gamma}}\\ &= 1 + \gamma h/2 + (\Omega^2 + \gamma^2 m)h^2/(4m) + \mathcal{O}(h^3).\end{aligned}$$

Hence we can see that the $[\![\mathrm{ABO}]\!]$ scheme gives a first-order error in the variance of q. For significantly small friction (relative to h), the leading order term will be dominated by the term at order h^2, potentially giving us greater accuracy than we would expect from a first order scheme.

However, we see that the error grows exponentially as γ gets large. We can explain this by returning to the definition of the scheme, which has an O step (stochastic fluctuation) immediately preceding the only A step (drift), wiping out all force information for infinite friction. In practice, playing with the friction coefficient can affect the rate at which averages converge towards the invariant average.

From the work in Chap. 3, we may expect that performing the integration steps symmetrically (by using a Strang splitting) could provide desirable higher order behavior. Computing the coefficients in (7.11) for the each of the symmetric